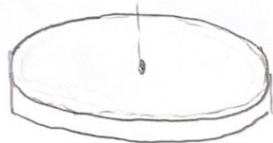


PHYS 302 HW2

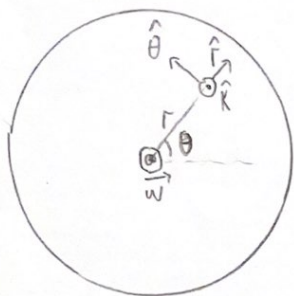
Boran Aybak Kiliç

(1)

g↓



⊗ g↓



$$\vec{a} = \vec{g} - 2 \vec{\omega} \times \vec{v} - \vec{\omega} \times (\vec{\omega} \times \vec{r}) + \frac{\vec{N}}{m}$$

* We should first derive the particles acceleration (of mass m) in cylindrical co-ordinates:

position vector $\vec{p} = r \hat{r} + z \hat{k}$

$$\vec{v} = \frac{d\vec{p}}{dt} = \dot{r} \hat{r} + r \dot{\theta} \hat{\theta} + \dot{z} \hat{k}$$

$$\vec{a} = \frac{d^2\vec{p}}{dt^2} = \ddot{r} \hat{r} + \dot{r} \dot{\theta} \hat{\theta} + \dot{r} \dot{\theta} \hat{\theta} + r \ddot{\theta} \hat{\theta} - r \dot{\theta}^2 \hat{r} + \ddot{z} \hat{k}$$

Note that we've used the basic properties of polar coordinates:

$$\frac{d\hat{r}}{dt} = \dot{\theta} \hat{\theta} \quad \text{and} \quad \frac{d\hat{\theta}}{dt} = -\dot{\theta} \hat{r}$$

These can be derived using the relation between derivatives of vectors on inertial and rotational frames. Namely $\left(\frac{d\vec{C}}{dt}\right)_{in} = \left(\frac{d\vec{C}}{dt}\right)_{rot} + \vec{\omega} \times \vec{C}$

$$\left(\frac{d\hat{r}}{dt}\right)_{in} = \left(\frac{d\hat{r}}{dt}\right)_{rot} + (\vec{\omega} \times \hat{r}) \Rightarrow \left(\frac{d\hat{r}}{dt}\right)_{in} = \dot{\theta} (\hat{k} \times \hat{r}) = \dot{\theta} \hat{\theta}$$

↳ $\left(\frac{d\hat{r}}{dt}\right)_{rot} = 0$ and $\vec{\omega} = \dot{\theta} \hat{k}$ since we're on a frame rotating with $\dot{\theta}$ and $\hat{r}$ will "seem" stagnant in this frame.

The other relation $\frac{d\hat{\theta}}{dt} = -\dot{\theta} \hat{r}$ can be showed using the same procedure.

We have $\vec{a} = (\ddot{r} - r \dot{\theta}^2) \hat{r} + (2\dot{r} \dot{\theta} + r \ddot{\theta}) \hat{\theta} + \ddot{z} \hat{k}$

and RHS of equation becomes:

$$\vec{a} = -g \hat{k} - 2\omega [\hat{k} \times (\dot{r} \hat{r} + r \dot{\theta} \hat{\theta} + \dot{z} \hat{k})] - \omega \hat{k} \times [\omega \hat{k} \times (r \hat{r} + z \hat{k})] + \frac{N}{m} \hat{k}$$

reaction force is perpendicular to the surface

$$\vec{a} = \left(\frac{N}{m} - g\right) \hat{k} - 2\omega \dot{r} \hat{\theta} + 2\omega r \dot{\theta} \hat{r} + \omega^2 r \hat{r}$$

Equating components of acceleration in two sides of the equation

$$\left. \begin{aligned} \ddot{r} - r\dot{\theta}^2 &= 2\omega r \dot{\theta} + \omega^2 r \\ 2\dot{r}\dot{\theta} + r\ddot{\theta} &= -2\omega \dot{r} \\ \ddot{z} &= \frac{N}{m} - g \end{aligned} \right\} \Rightarrow \begin{aligned} \ddot{r} &= r\dot{\theta}^2 + 2\omega r \dot{\theta} + \omega^2 r & (\hat{r}) \\ 2\dot{r}\dot{\theta} + r\ddot{\theta} &= -2\omega \dot{r} & (\hat{\theta}) \\ \ddot{z} &= \frac{N}{m} - g = 0 & (\hat{k}) \end{aligned}$$

rest along vertical

Now, let $r^2\dot{\theta} + r^2\omega = \gamma$ (γ some scalar quantity)

$$\frac{d}{dt}(r^2\dot{\theta} + r^2\omega) = 2r\dot{r}\dot{\theta} + r^2\ddot{\theta} + 2r\dot{r}\omega + r^2 \underbrace{\frac{d\omega}{dt}}_0 \rightarrow \text{since } \vec{\omega} \text{ constant of this motion}$$

$$= r(2\dot{r}\dot{\theta} + r\ddot{\theta} + 2\dot{r}\omega) = \frac{d\gamma}{dt} = 0$$

this can easily be seen from above equation for $\hat{\theta}$

Then since $\frac{d\gamma}{dt} = 0$, $\underline{r^2\dot{\theta} + r^2\omega = \gamma}$ is constant of this motion //

Say $r^2\dot{\theta} + r^2\omega = l^2\Omega$ where $l^2\Omega \equiv r^2(0)\dot{\theta}(0) + r^2(0)\omega$

From $\hat{r}$ equation $\ddot{r} = r(\dot{\theta}^2 + 2\omega\dot{\theta} + \omega^2) = r(\dot{\theta} + \omega)^2$

$$\ddot{r} = r \left(\frac{l^2\Omega}{r^2} \right)^2 = \frac{l^4\Omega^2}{r^3} //$$

$$2\dot{r}\ddot{r} - 2\dot{r} \frac{l^4\Omega^2}{r^3} = 0 \Rightarrow \frac{d}{dt} \left(\dot{r}^2 + \frac{l^4\Omega^2}{r^2} \right) = 0$$

This can be seen by taking the derivative of the given expression as well.

$\Rightarrow$ Then $\dot{r}^2 + \frac{l^4\Omega^2}{r^2} = A$ is constant of this motion //

Let $\dot{r}^2 + \frac{l^4 \Omega^2}{r^2} = A$ (some constant)

$$\dot{r}^2 = \frac{Ar^2 - l^4 \Omega^2}{r^2} \quad ; \quad \dot{r} = \pm \frac{\sqrt{Ar^2 - l^4 \Omega^2}}{r}$$

$$\pm \frac{r dr}{\sqrt{Ar^2 - l^4 \Omega^2}} = dt \xrightarrow{\substack{Ar^2 - l^4 \Omega^2 = u \\ 2A r dr = du}} \pm \int_{u(0)}^{u(t)} \frac{du}{2A\sqrt{u}} = \pm \int_0^t dt$$

$$\pm \sqrt{Ar^2 - l^4 \Omega^2} \Big|_{r(0)}^{r(t)} = \sqrt{Ar^2 - l^4 \Omega^2} - \sqrt{Ar(0)^2 - l^4 \Omega^2} = \pm At$$

Since A is constant pick $A = \dot{r}(0)^2 + \frac{l^4 \Omega^2}{r(0)^2}$

$$\sqrt{Ar^2 - l^4 \Omega^2} - \sqrt{r(0)^2 \dot{r}(0)^2 + l^4 \Omega^2 - l^4 \Omega^2} = \pm At \quad (\text{given that } r(0) \neq 0)$$

$$Ar^2 - l^4 \Omega^2 = (r(0) \dot{r}(0) \pm At)^2 \quad ; \quad (Ar^2 = r^2 \dot{r}^2 + l^4 \Omega^2)$$

$$(r \dot{r})^2 = (r(0) \dot{r}(0) \pm At)^2 \Rightarrow \left(\frac{1}{2} \frac{d}{dt} r^2 \right)^2 = (r(0) \dot{r}(0) \pm At)^2$$

$$\frac{1}{2} \frac{dr^2}{dt} = \pm (r(0) \dot{r}(0) \pm At) \Rightarrow \frac{1}{2} (r^2 - r^2(0)) = \left(\pm r(0) \dot{r}(0) \pm A \frac{t^2}{2} \right)$$

$$r^2 = \pm 2 r(0) \dot{r}(0) t + A t^2 + r^2(0)$$

$$r^2 = r^2(0) \pm 2 r(0) \dot{r}(0) t + \dot{r}^2(0) t^2 + \frac{l^4 \Omega^2}{r^2(0)} t^2$$

$$r^2 = (r(0) \pm \dot{r}(0) t)^2 + \frac{l^4 \Omega^2}{r^2(0)} t^2$$

Now we have :

$$r(t)^2 = (r(0) \pm \dot{r}(0)t)^2 + \frac{\ell^2 \Omega^2}{(\dot{\theta}(0) + \omega)^2} t^2 \Rightarrow$$

This is in some ambiguous
resemblance with 1-D
linear motion!

$$r(t) = \pm \sqrt{(r(0) \pm \dot{r}(0)t)^2 + (\dot{\theta}(0) + \omega)^2 t^2}$$

$$A r(0) = A^2 c_1^2 + \ell^2 \Omega^2$$

$$c_1 = \pm \frac{\dot{r}(0) r(0)}{A}$$

or in terms of $\ell^2 \Omega^2$ and A (constant of this motion):

$$r(t) = \pm \frac{\sqrt{\pm 2 A^2 c_1 t + A^2 c_1^2 + A^2 t^2 + \ell^2 \Omega^2}}{\sqrt{A}}$$

c_1 is to be found
 $\Rightarrow$ using initial conditions

Now to find $\theta(t)$ use, $(\dot{\theta}(t) + \omega) \times \left(\pm 2 A^2 c_1 t + A^2 c_1^2 + A^2 t^2 + \frac{\ell^2 \Omega^2}{A} \right) = \ell^2 \Omega^2$

Using Mathematica one can find $\theta(t)$ as:

$$\theta(t) = \frac{\ell^2 \Omega \tan^{-1} \left(\frac{A(t - c_1)}{\ell^2 \Omega} \right) + k_1 - \omega t}{\ell^2 \Omega}$$

$$\theta(t) = \tan^{-1} \left(\frac{A t}{\ell^2 \Omega} \pm \frac{\dot{r}(0) r(0)}{\ell^2 \Omega} \right) + k_1 - \omega t$$

$$\theta(0) = \tan^{-1} \left(\pm \frac{\dot{r}(0) r(0)}{\ell^2 \Omega} \right) + k_1 \Rightarrow k_1 = \pm \tan^{-1} \left(\frac{\dot{r}(0) r(0)}{\ell^2 \Omega} \right) + \theta(0)$$

$$\theta(t) = \tan^{-1} \left(\frac{A t \pm \dot{r}(0) r(0)}{\ell^2 \Omega} \right) \mp \tan^{-1} \left(\frac{\dot{r}(0) r(0)}{\ell^2 \Omega} \right) + \theta(0) - \omega t //$$

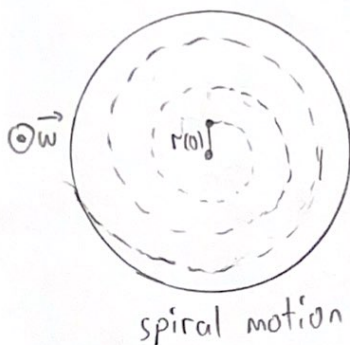
2) $\dot{\theta}(0) = -\omega$, then $\dot{\theta}^2 \Omega = 0$ [by $\dot{\theta}^2 \Omega = r^2(0)(\dot{\theta}(0) + \omega)$]

$\dot{r}^2 = A$ constant $\Rightarrow \dot{r} = \sqrt{A}$ constant

$r(t) = \sqrt{A}t + r(0) = \dot{r}(0)t + r(0)$

$\dot{\theta} + \omega = 0 \Rightarrow \theta(t) = -\omega t + \theta(0)$

From rotating observers perspective:



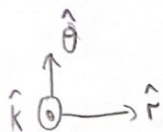
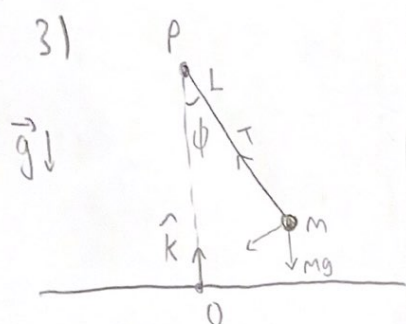
* In rotating frame object is going to rotate with constant $-\omega$ while getting further away from the center by a linear function of t . Object is going to spiral out/in from/to center depending on the sign of $\dot{r}(0)$.

* If $\dot{r}(0) = 0$, distance from the center is constant and it will draw circles around the center.

From inertial observers perspective:

Net angular speed $\vec{\omega} + \dot{\theta}(t)\hat{k}$ is 0. Thus object will not 'rotate' at all. It will just move radially out/in depending on the sign of $\dot{r}(0)$ with a constant speed.

* If $\dot{r}(0) = 0$, object will appear completely stationary to the inertial observer.

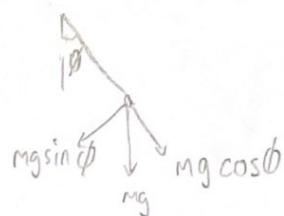


This solution is done using this set of cylindrical coord.

In inertial frame using cylindrical coordinates.

$$\vec{a} = \frac{\vec{T}}{m} + \vec{g}$$

$$\vec{a} = -\frac{T}{m} \hat{r} + g \cos \phi \hat{r} - g \sin \phi \hat{\phi}$$



We have derived the expression for acceleration in cylindrical coordinates in problem (1) as:

$$\vec{a} = (\ddot{r} - r\dot{\phi}^2) \hat{r} + (2\dot{r}\dot{\phi} + r\ddot{\phi}) \hat{\phi} + \ddot{z} \hat{k}$$

$$a_r = \ddot{r} - r\dot{\phi}^2 = g \cos \phi - \frac{T}{m}$$

$$\frac{T}{m} = g \cos \phi + r\dot{\phi}^2$$

$\Rightarrow$ Since rope or the rod that the ball is attached is not elastic, distance of the ball to the fixed point will be constant. That is $r = L$, $\dot{r} = \ddot{r} = 0$.

$a_z = \ddot{z} = 0 \Rightarrow$ there is no force perpendicular to the plane of oscillations in the inertial frame. Thus motion must be planar in $\hat{r}\hat{k}$ plane

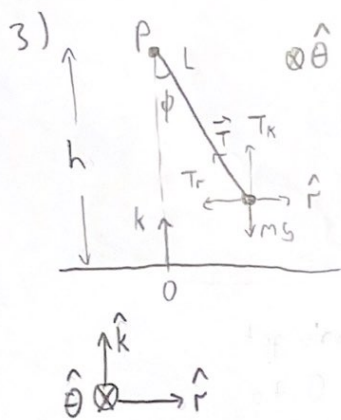
$$a_\phi = 2\dot{r}\dot{\phi} + r\ddot{\phi} = -g \sin \phi \Rightarrow \text{Since } r = L \text{ and } \dot{r} = 0$$

$$\ddot{\phi} + \frac{g}{L} \sin \phi = 0$$

If we assume ϕ is small, $\sin \phi \approx \phi$

$$\ddot{\phi} + \frac{g}{L} \phi = 0 \Rightarrow \underline{\phi(t) = A \sin(\omega t) + B \cos(\omega t)} \quad \text{with } \omega = \sqrt{\frac{g}{L}}$$

$\hookrightarrow$ simple harmonic oscillation motion

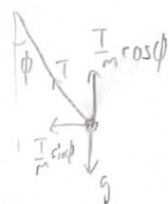


This solution is done using this set of cylindrical coordinates

In inertial frame using cylindrical coordinates:

$$\vec{a} = \frac{\vec{T}}{m} + \vec{g}$$

$$\vec{a} = + \frac{T}{m} \cos \phi \hat{k} - \frac{T}{m} \sin \phi \hat{r} - g \hat{k}$$



We have derived the expression for acceleration in cylindrical co-ordinates in problem (1) as:

$$\vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{r} + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta} + \ddot{z}\hat{k}$$

$a_{\hat{\theta}} = 2\dot{r}\dot{\theta} + r\ddot{\theta} = 0 \Rightarrow$ There is no force in $\hat{\theta}$ direction. Thus it is obvious that $\dot{\theta}$ and $\ddot{\theta}$ equals to zero since r and $\dot{r}$ are arbitrarily nonzero. Then if θ is constant motion is planar

$$a_z = \ddot{z} = \frac{T}{m} \cos \phi - g \quad \text{and} \quad a_r = \ddot{r} - \underbrace{r\dot{\theta}^2}_0 = \ddot{r} = -\frac{T}{m} \sin \phi$$

From the figure above notice $z = h - L \cos \phi$ and $r = L \sin \phi$

$$\dot{z} = L \sin \phi \frac{d\phi}{dt}, \quad \dot{r} = L \cos \phi \frac{d\phi}{dt}$$

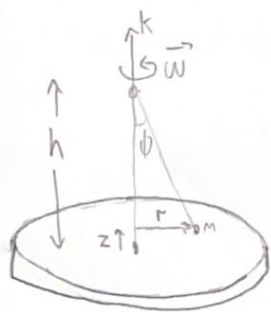
$$\ddot{z} = L \left[\cos \phi \left(\frac{d\phi}{dt} \right)^2 + \sin \phi \frac{d^2\phi}{dt^2} \right], \quad \ddot{r} = L \left[-\sin \phi \left(\frac{d\phi}{dt} \right)^2 + \cos \phi \frac{d^2\phi}{dt^2} \right]$$

$$(\ddot{z} + g) \sin \phi = -\ddot{r} \cos \phi = \frac{T}{m} \cos \phi \sin \phi \quad \left(\begin{array}{l} \text{from expressions} \\ \text{for } \ddot{z} \text{ and } \ddot{r} \end{array} \right)$$

$$L \sin \phi \cos \phi \left(\frac{d\phi}{dt} \right)^2 + L \sin^2 \phi \frac{d^2\phi}{dt^2} + g \sin \phi = L \sin \phi \cos \phi \left(\frac{d\phi}{dt} \right)^2 - L \cos^2 \phi \frac{d^2\phi}{dt^2}$$

$$\frac{d^2\phi}{dt^2} + \frac{g}{L} \sin \phi = 0 \Rightarrow \left(\begin{array}{l} \ddot{\phi} + \frac{g}{L} \phi = 0 \\ \text{(assuming } \phi \text{ small)} \end{array} \right) \Rightarrow \phi = A \sin \omega t + B \cos \omega t \quad \text{where } \omega = \sqrt{\frac{g}{L}}$$

4)



Now since the pendulum system is rotating with $\vec{\omega}$, it would be pertinent to study this system in the rotating frame

$$\vec{a}_{\text{rot}} = \frac{\vec{T}}{m} + g - 2(\vec{\omega} \times \vec{v}) - \vec{\omega} \times (\vec{\omega} \times \vec{r})$$

$$\vec{r} = r\hat{r} + z\hat{k}, \quad \vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} + \dot{z}\hat{k}$$

$$\vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{r} + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta} + \ddot{z}\hat{k}$$

* We are constraining our study for the motion of the ball for small ϕ . This means r and z will always be small compared to $\dot{r}, \ddot{r}, \dot{\theta}, \ddot{\theta}$ etc. Thus we can make the approximation $r \approx 0$. But let us first clearly write the full equation of motion

$$\vec{a} = \left(\frac{T}{m} \cos \phi - g \right) \hat{k} + \left(-T \sin \phi \hat{r} \right) - 2 \left[\omega \hat{k} \times (\dot{r}\hat{r} + r\dot{\theta}\hat{\theta} + \dot{z}\hat{k}) \right] - \omega \hat{k} \times [\omega \hat{k} \times (r\hat{r} + z\hat{k})]$$

$$= \left(\frac{T}{m} \cos \phi - g + \omega^2 r \right) \hat{k} + (2\omega r\dot{\theta} - T \sin \phi) \hat{r} + (-2\omega \dot{r}) \hat{\theta}$$

$$\left. \begin{aligned} \ddot{r} - r\dot{\theta}^2 &= 2\omega r\dot{\theta} - T \sin \phi \\ 2\dot{r}\dot{\theta} + r\ddot{\theta} &= -2\omega \dot{r} \\ \ddot{z} &= \frac{T}{m} \cos \phi - g + \omega^2 r \end{aligned} \right\} \approx \begin{aligned} \ddot{r} &= -T \sin \phi & (\hat{r} \text{ equation}) \\ 2\dot{r}\dot{\theta} &= -2\omega \dot{r} & (\hat{\theta} \text{ equation}) \\ \ddot{z} &= \frac{T}{m} \cos \phi - g & (\hat{k} \text{ equation}) \end{aligned}$$

* We could have instead leave r as it is and make small angle approximations in later steps. Nevertheless, it would yield the same result.

We now know that $\Rightarrow \ddot{r} = -T \sin \phi$, $\ddot{z} = \frac{T}{m} \cos \phi - g$, $\dot{\theta} = -\omega$

This result is very similar to the last question where pendulum was not rotating. The only difference is now the plane of oscillations is not fixed but rather it is rotating with $-\omega$. Thus trajectory of the ball is governed by two motions: harmonic oscillations in a fixed plane and the rotation of that plane which are independent from each other as long as $r \neq 0$ or $\phi \neq 0$.

$$\phi = A \sin(\omega_0 t) + B \cos(\omega_0 t) \quad \text{where } \omega_0 = \sqrt{\frac{g}{L}}$$

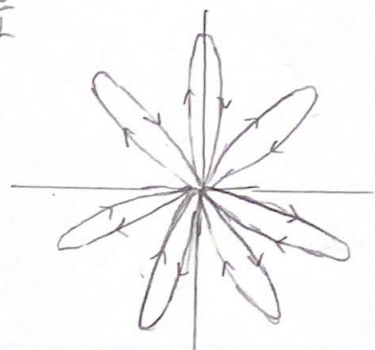
A and B can be determined from initial conditions

$$r(t) = L \sin \phi$$

$$z(t) = h - L \cos \phi$$

$$\theta(t) = -\omega t + \theta(0)$$

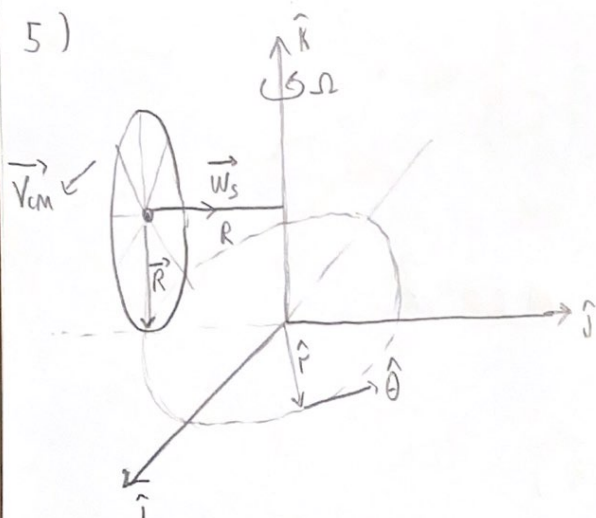
} describes the full motion of the ball



* A and B constant are determined by the initial conditions.

trajectory of the ball seen from above

5)



a) We can separate the motion of the hoop as rotation around its symmetry axis with angular speed $\vec{\omega}$ and rotation around $\hat{k}$ with $\vec{\Omega}$

→ Notice the hoop is rotating without slipping with constant angular freq., it necessitates the fact that there must be a torque acting on the hoop. Moreover direction of $\vec{L}$ will change (following a conic path), so it's sensible to study this system in cylindrical coordinates

* Condition for rolling without slipping;

$$\vec{V}_{\text{contact}} = 0$$

$$\vec{V}_{\text{contact}} = \underbrace{\vec{\Omega} \times \vec{r}_{\text{cm}}}_{\vec{V}_{\text{cm}}} + \vec{\omega}_s \times \vec{R} = 0$$

$$= \Omega R \hat{\theta} - \omega_s R \hat{\theta} = 0 \Rightarrow \omega_s = \Omega$$

$$\vec{\omega} = \vec{\omega}_s + \vec{\Omega} = \boxed{\Omega (-\hat{r} + \hat{k})} \quad \text{OR} \quad \vec{\omega} = \Omega (\hat{j} + \hat{k})$$

↗ If one were to observe this system at the instant sketched above.

b) In our discussion in class we've found that total angular momentum of the system can be attributed to the sum of the angular momentum of the CM w.r.t. the origin (as if it's a point particle) and angular momentum of the object w.r.t. its CM. (spin angular momentum).

$$\vec{L}_T = M_T \vec{r}_{\text{cm}} \times \vec{V}_{\text{cm}} + \sum_{i=1}^N m_i \vec{p}_i \times \frac{d\vec{p}_i}{dt} \Rightarrow (\vec{L}_T)_a = M_T (\vec{r}_{\text{cm}} \times \vec{V}_{\text{cm}})_a + I_{ab} \omega_b \quad (\text{sum over } b)$$

Since the symmetry axes for the hoop are also principal axes, moment of inertia tensor (w.r.p CM) is diagonal }
$$I = \begin{bmatrix} \frac{1}{2}MR^2 & 0 & 0 \\ 0 & MR^2 & 0 \\ 0 & 0 & \frac{1}{2}MR^2 \end{bmatrix}$$

This can be calculated using: $I_{ab} = \sum_{i=1}^N [m_i p_i^2 \delta_{ab} - m_i (p_i)_a (p_i)_b]$

One finds $I_{ab} = 0$ for $a \neq b$ in $\hat{i}, \hat{j}, \hat{k}$ coordinates

* OR equivalently in $\hat{\theta}, \hat{r}, \hat{k}$ coordinates since they are orthonormal unit vectors.

$$\vec{L}_{CM}^{of} = \begin{bmatrix} \frac{1}{2}MR^2 & 0 & 0 \\ 0 & MR^2 & 0 \\ 0 & 0 & \frac{1}{2}MR^2 \end{bmatrix} \begin{bmatrix} 0 \\ -\Omega \\ \Omega \end{bmatrix} = -MR^2\Omega \hat{r} + \frac{1}{2}MR^2\hat{k}$$

↳ total angular momentum w.r.t. CM

$$\vec{L}_{CM} = M_T(\vec{r}_{CM} \times \vec{v}_{CM}) = M(R\hat{r} \times \Omega R\hat{\theta}) = MR^2\Omega\hat{k}$$

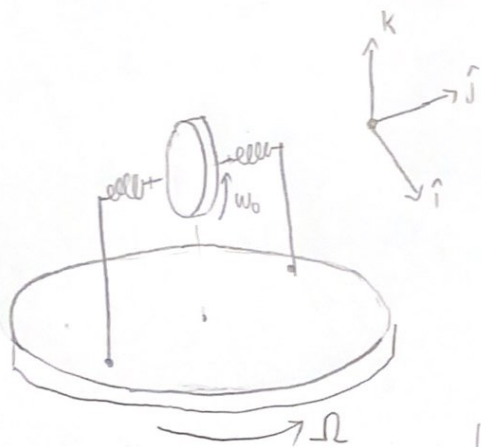
$$\vec{L}_T = \vec{L}_{CM} + \vec{L}_{CM}^{of} = \Omega MR^2(-\hat{r} + \frac{3}{2}\hat{k})$$

If one were to calculate it at instant given sketch:

$$\vec{L}_T = \Omega MR^2(\hat{j} + \frac{3}{2}\hat{k})$$

→ Clearly $\vec{L}_T$ and $\vec{\omega}$ are not parallel.

6)



* Let us first study the motion in the rotating frame whose axes instantaneously coincide with the symmetry axes of the object,

$$\left(\frac{d\vec{L}}{dt}\right)_{in} = \vec{\tau} = \underbrace{\left(\frac{d\vec{L}}{dt}\right)_{rot}}_{\text{frame rotating with } \Omega} + \vec{\omega} \times \vec{L}$$

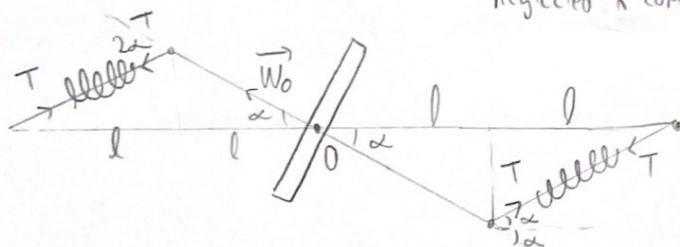
$$\left(\frac{d\vec{L}}{dt}\right)_{rot} = 0$$

$$\vec{\tau} = (\Omega \hat{k}) \times (L_0 \hat{j}) = \Omega L_0 \hat{i} \Rightarrow$$

neglected $\hat{k}$ component

There will be a torque in $\hat{i}$ direction so axis of the flywheel will be tilted w.r.t. horizontal.

↓
forced precession



Now again when we write $\vec{\tau} = \vec{\omega} \times \vec{L} = \left(\frac{d\vec{L}}{dt}\right)_{in}$

$$(\vec{L})_c = I_{ab} \vec{\omega}_b \Rightarrow \vec{L} = \begin{bmatrix} \frac{1}{4}MR^2 & 0 & 0 \\ 0 & \frac{1}{2}MR^2 & 0 \\ 0 & 0 & \frac{1}{4}MR^2 \end{bmatrix} \begin{bmatrix} 0 \\ -\omega_0 \sin \alpha \\ \omega_0 \cos \alpha + \Omega \end{bmatrix}$$

$$\vec{L} = -L_0 \sin \alpha \hat{j} + (L_0 \cos \alpha + L_\Omega) \hat{k} \quad \text{where } L_0 = I_0 \omega_0 = \frac{1}{2}MR^2 \omega_0$$

$$L_\Omega = \frac{1}{4}MR^2 \Omega$$

$$\vec{\tau} = \Omega \hat{k} \times [-L_0 \sin \alpha \hat{j} + (L_0 \cos \alpha + L_\Omega) \hat{k}]$$

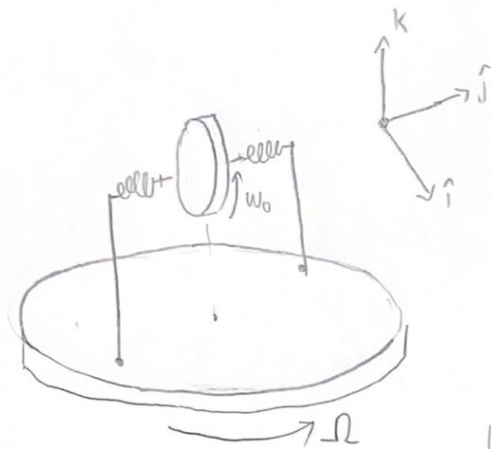
$$\vec{\tau} = +\Omega L_0 \hat{i}$$

since α small

* Torque with respect to CM of flywheel: $\vec{\tau} = 2Tl \sin 2\alpha \hat{i} \approx 4Tl \alpha \hat{i}$

$$\vec{\tau} = \Omega L_0 \hat{i} = 4Tl \alpha \hat{i} \Rightarrow \boxed{\alpha = \frac{\Omega L_0}{4Tl} = \frac{\Omega MR^2 \omega_0}{8Tl}}$$

6)



* Let us first study the motion in the rotating frame whose axes instantaneously coincide with the symmetry axes of the object,

$$\left(\frac{d\vec{L}}{dt}\right)_{in} = \vec{\tau} = \underbrace{\left(\frac{d\vec{L}}{dt}\right)_{rot}}_{\text{frame rotating with } \Omega} + \vec{\omega} \times \vec{L}$$

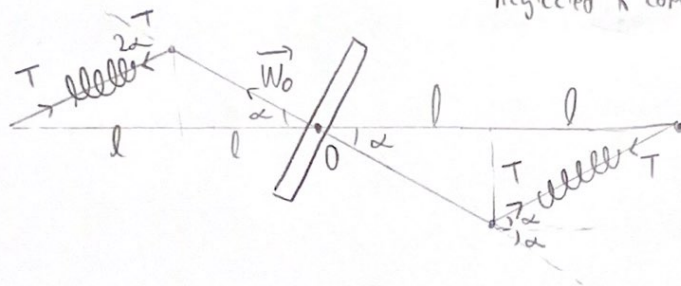
$$\left(\frac{d\vec{L}}{dt}\right)_{rot} = 0$$

$$\vec{\tau} = (\Omega \hat{k}) \times (L_0 \hat{j}) = \Omega L_0 \hat{i} \Rightarrow$$

neglected $\hat{k}$ component

There will be a torque in $\hat{i}$ direction so axis of the flywheel will be tilted w.r.t. horizontal.

↓
forced precession,



Now again when we write $\vec{\tau} = \vec{\omega} \times \vec{L} = \left(\frac{d\vec{L}}{dt}\right)_{in}$

$$(\vec{L})_c = I_{ab} \vec{\omega}_b \Rightarrow \vec{L} = \begin{bmatrix} \frac{1}{4}MR^2 & 0 & 0 \\ 0 & \frac{1}{2}MR^2 & 0 \\ 0 & 0 & \frac{1}{4}MR^2 \end{bmatrix} \begin{bmatrix} 0 \\ -\omega_0 \sin \alpha \\ \omega_0 \cos \alpha + \Omega \end{bmatrix}$$

$$L = -L_0 \sin \alpha \hat{j} + (L_0 \cos \alpha + L_\Omega) \hat{k} \quad \text{where } L_0 = I_0 \omega_0 = \frac{1}{2}MR^2 \omega_0$$

$$L_\Omega = \frac{1}{4}MR^2 \Omega$$

$$\vec{\tau} = \Omega \hat{k} \times [-L_0 \sin \alpha \hat{j} + (L_0 \cos \alpha + L_\Omega) \hat{k}]$$

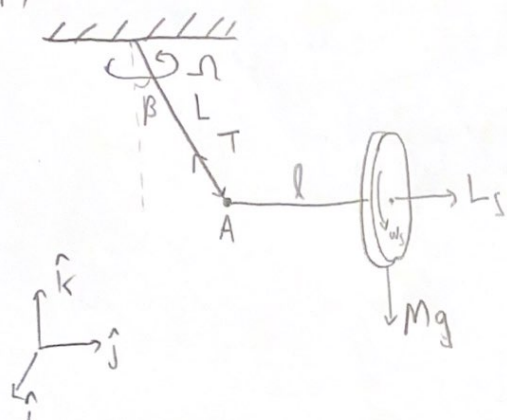
$$\vec{\tau} = +\Omega L_0 \hat{i}$$

since α small

* Torque with respect to CM of flywheel: $\vec{\tau} = 2Tl \sin 2\alpha \hat{i} \approx 4Tl\alpha \hat{i}$

$$\vec{\tau} = \Omega L_0 \hat{i} = 4Tl\alpha \hat{i} \Rightarrow \alpha = \frac{\Omega L_0}{4Tl} = \frac{\Omega MR^2 \omega_0}{8Tl}$$

7)



* If motion of the gyroscope is governed by uniform precession in the horizontal plane, we must have the total acceleration in the rotating frame rotating with Ω (angular speed of precession) equal to 0.

$$\vec{a}_{\text{rot}} = \frac{\vec{T}}{m} + \vec{g} - 2(\vec{\Omega} \times \vec{v}_{\text{rot}}) - \vec{\Omega} \times (\vec{\Omega} \times \vec{r})$$

At the sketched instant above (we can choose to observe this system at any instant because of the uniformity of the motion without loss of generality)

$$\vec{\Omega} = \Omega \hat{k}, \quad \vec{v}_{\text{rot}} = 0 \quad (\text{rotating frame})$$

$$\vec{r} = (l + L \sin \beta) \hat{j}$$

$$(\vec{a})_{\text{rot}} = 0 = \left(\frac{T \cos \beta}{M} - g \right) \hat{k} + \left(-\Omega^2 (l + L \sin \beta) - \frac{T \sin \beta}{M} \right) \hat{j}$$

With the small angle approximation:

$$Mg = T \cos \beta \approx T, \quad \Omega^2 = \frac{T \sin \beta}{L \sin \beta + l} \approx \frac{T \beta}{(l + L \beta) M} = \frac{g \beta}{l + L \beta}$$

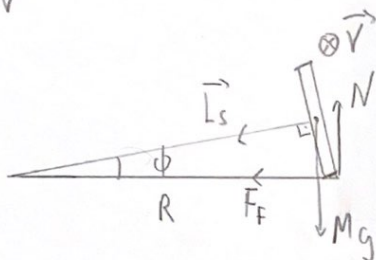
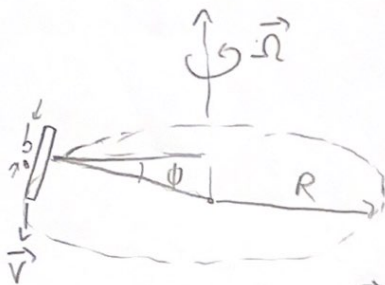
Now from our previous discussion it can easily be seen that,

$$\vec{\tau}_A = \frac{d\vec{L}_s}{dt} = L_s \Omega (-\hat{i}) = l Mg (-\hat{i}) \Rightarrow \Omega = \frac{Mg l}{L_s} \rightarrow L_s = I_0 \omega_s \quad \text{spin angular mom.}$$

$$\Omega^2 = \frac{M^2 g^2 l^2}{I_0^2 \omega_s^2} = \frac{g \beta}{l + L \beta} \Rightarrow \beta \left(1 - \frac{M^2 g^2 l^2 L}{I_0^2 \omega_s^2} \right) = \frac{M^2 g^2 l^3}{I_0^2 \omega_s^2}$$

$$\boxed{\beta = \frac{M^2 g^2 l^3}{I_0^2 \omega_s^2 - M^2 g^2 l^2 L}}$$

8)



* This is again a problem of forced precession in the presence of external torques.

In the rotating frame, rotating with Ω there will be no acceleration of the CM of coin.

$$(\vec{a})_{\text{rot}} = \frac{\vec{N}}{M} + \frac{\vec{F}_f}{M} + \vec{g} - 2(\vec{\Omega} \times \vec{V}_{\text{rot}}) - \vec{\Omega} \times (\vec{\Omega} \times \vec{r})$$

$$\vec{V}_{\text{rot}} = 0, \quad \vec{\Omega} = \Omega \hat{k}, \quad \vec{N} = N \hat{j}, \quad \vec{F}_f = -F_f \hat{j}, \quad \vec{g} = -g \hat{j}$$

* Assume for the rest of this problem CM point is almost distance R away from the center although in reality it's a bit less.

$$(\vec{a})_{\text{rot}} = 0 = \left(\frac{N}{M} - g\right) \hat{j} + (\Omega^2 R - \frac{F_f}{M}) \hat{j} \Rightarrow \begin{aligned} N &= Mg \\ F_f &= M \Omega^2 R \end{aligned}$$

$$\text{In this rotating frame } \left(\frac{d\vec{L}_s}{dt}\right)_{\text{rot}} = 0$$

$$\left(\frac{d\vec{L}_s}{dt}\right)_{\text{in}} = \vec{\tau} = \vec{\Omega} \times \vec{L}_s$$

$\Rightarrow$ we only consider spin angular momentum because angular momentum due to $\vec{\Omega}$ will be constant throughout the motion and will not cause a torque

$$\vec{\tau} = \Omega \hat{k} \times (L_s \cos \phi (-\hat{j}) + L_s \sin \phi (\hat{k})) = \Omega L_s \cos \phi \hat{i}$$

$$\vec{\tau}_{\text{cm}} = (N b \sin \phi - F_f b \cos \phi) \hat{i} = \Omega L_s \cos \phi \hat{i}$$

* Condition for rolling without slipping $\Rightarrow V_{\text{contact}} = 0$

$$\vec{V}_{\text{contact}} = \vec{V}_{\text{cm}} + \vec{b} \times \vec{\omega}_s \Rightarrow \Omega R - b \omega_s = 0 \Rightarrow V = \Omega R = b \omega_s$$

$$\text{Also } L_s = I_0 \omega_s = \frac{1}{2} M b^2 \omega_s$$

$$\tau_{cm} = M g b \sin \phi - M \Omega^2 R b \cos \phi = \Omega \frac{1}{2} M b^2 \omega_s \cos \phi$$

$$M g b \sin \phi - M \frac{v^2}{R} b \cos \phi = \frac{1}{2} M \frac{v^2}{R} b \cos \phi$$

$$\tan \phi = \frac{3v^2}{2Rg}$$